

(4)

$$\frac{J(x_1, x_2, \dots, x_n, b + \Delta b) - J(x_1, x_2, \dots, x_n, b)}{\Delta b} \quad (\Delta b \neq 0)$$

$$= \frac{1}{\Delta b} \cdot (J(\vec{x}, b + \Delta b) - J(\vec{x}, b))$$

$$= \frac{1}{\Delta b} \cdot \left[\frac{1}{n} \cdot \sum_{i=1}^n \left(f(x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}) - y^{(i)} \right)^2 \right. \\ \left. - \frac{1}{n} \cdot \sum_{i=1}^n \left(f(x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}) - y^{(i)} \right)^2 \right]$$

$$= \frac{1}{n \cdot \Delta b} \cdot \left[\sum_{i=1}^n \left(f(\vec{x}^{(i)}) - y^{(i)} \right)^2 - \sum_{i=1}^n \left(f_b(\vec{x}^{(i)}) - y^{(i)} \right)^2 \right]$$

$$= \frac{1}{n \cdot \Delta b} \cdot \sum_{i=1}^n \left[\underline{\underline{\left(f_{b+\Delta b}(\vec{x}^{(i)}) - y^{(i)} \right)^2}} - \underline{\underline{\left(f_b(\vec{x}^{(i)}) - y^{(i)} \right)^2}} \right]$$

$$= \frac{1}{n \cdot \Delta b} \cdot \sum_{i=1}^n \left[\left(2x_1^{(i)} + 2x_2^{(i)} + \dots + 2x_n^{(i)} + \underline{b + \Delta b} - y^{(i)} \right)^2 \right. \\ \left. - \left(2x_1^{(i)} + 2x_2^{(i)} + \dots + 2x_n^{(i)} + \underline{b} - y^{(i)} \right)^2 \right]$$

$$= \frac{1}{n \cdot \Delta b} \cdot \sum_{i=1}^n \Delta b \cdot \left(2x_1^{(i)} \cdot 2 + 2x_2^{(i)} \cdot 2 + \dots + 2x_n^{(i)} \cdot 2 + 2b - 2y^{(i)} + \Delta b \right)$$

$$= \frac{1}{n} \cdot \sum_{i=1}^n \left(2x_1^{(i)} \cdot 2 + 2x_2^{(i)} \cdot 2 + \dots + 2x_n^{(i)} \cdot 2 + 2b - 2y^{(i)} + \Delta b \right)$$

~~$$= \frac{1}{n} \cdot \sum_{i=1}^n \left(2x_1^{(i)} \cdot 2 + 2x_2^{(i)} \cdot 2 + \dots + 2x_n^{(i)} \cdot 2 + 2b - 2y^{(i)} + \Delta b \right)$$~~

(5)

$$= \frac{1}{n} \cdot \left[\sum_{i=1}^n (2 \cdot x_1^{(i)} \cdot z_1 + 2 \cdot x_2^{(i)} \cdot z_2 + \dots + 2 \cdot x_n^{(i)} \cdot z_n + 2b - 2 \cdot y^{(i)}) + \sum_{i=1}^n \Delta b \right]$$

$$= \frac{1}{n} \cdot \left[\sum_{i=1}^n (2 \cdot x_1^{(i)} \cdot z_1 + 2 \cdot x_2^{(i)} \cdot z_2 + \dots + 2 \cdot x_n^{(i)} \cdot z_n + 2b - 2 \cdot y^{(i)}) + n \cdot \Delta b \right]$$

$$\left[\frac{J(\vec{a}, b + \Delta b) - J(\vec{a}, b)}{\Delta b} \right] = \left[\frac{1}{n} \cdot \sum_{i=1}^n 2 \cdot (f(\vec{x}^{(i)}) - y^{(i)}) + \boxed{\Delta b} \right]$$

$\Delta b \rightarrow 0$
↓
0

$\Delta b \rightarrow 0$
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$$\left[\frac{1}{n} \cdot \sum_{i=1}^n 2 \cdot (f(\vec{x}^{(i)}) - y^{(i)}) \right]$$

$$\left[\frac{\partial J(a_1, a_2, \dots, a_n, b)}{\partial b} \right] = \left[\frac{2}{n} \cdot \sum_{i=1}^n (f(x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}) - y^{(i)}) \right]$$

||

$$\left[\frac{2}{n} \cdot \sum_{i=1}^n (f(\vec{x}^{(i)}) - y^{(i)}) \right]$$